

When a particle moves in 3-space, its path is passing through various (x, y, z) -coordinates. If we let each coordinate become dependent on a parameter t , then the path of the particle can become parametrized by $C(x, y, z) = (f(t), g(t), h(t))$ where t lives on some domain I . If we let each coordinate live at the endpoint P of a vector $\mathbf{r} = \overrightarrow{OP}$, where O is the origin, then this curve can be written as the following vector:

$$\mathbf{r}(t) = \overrightarrow{OP} = f(t)\mathbf{i} + g(t)\mathbf{j} + h(t)\mathbf{k} = \langle f(t), g(t), h(t) \rangle$$

This object is known as a **vector-valued** function, contrasting with the *scalar-valued* functions we studied in single-variable calculus. Largely speaking, everything we want from single-variable calculus translates perfectly as long as we choose to consider $\mathbf{r}(t)$ component-wise. For example:

Definition. Given the vector-function $\mathbf{r}(t) = \langle f(t), g(t), h(t) \rangle$, if L_1 , L_2 , and L_3 are each the (finite) limits of $f(t)$, $g(t)$, and $h(t)$, respectively, at t_0 , then

$$\lim_{t \rightarrow t_0} \mathbf{r}(t) = \mathbf{L}$$

where $\mathbf{L} = \langle L_1, L_2, L_3 \rangle$.

Thus, $\mathbf{r}(t)$ is continuous at $t = t_0$ if $\lim_{t \rightarrow t_0} \mathbf{r}(t) = \mathbf{r}(t_0)$. (i.e., component-wise continuity)

Example. Graph the helix $\mathbf{r}(t) = \langle \cos(t), \sin(t), t \rangle$, where $-\infty < t < \infty$, and compute $\lim_{t \rightarrow \frac{\pi}{4}} \mathbf{r}(t)$.

Definition. The vector-function $\mathbf{r}(t) = \langle f(t), g(t), h(t) \rangle$ is differentiable at t if f , g , and h all have derivatives at t . Additionally,

$$\mathbf{r}'(t) = \frac{d\mathbf{r}}{dt} = \lim_{\Delta t \rightarrow 0} \frac{\mathbf{r}(t + \Delta t) - \mathbf{r}(t)}{\Delta t} = \left\langle \frac{df}{dt}, \frac{dg}{dt}, \frac{dh}{dt} \right\rangle = \langle f'(t), g'(t), h'(t) \rangle$$

The derivative $\mathbf{r}'(t)$ *always points in the direction of motion*.

The curve traced out by $\mathbf{r}(t)$ is **smooth** if $\mathbf{r}'(t)$ is continuous and never equal to the zero-vector $\vec{0}$. Equivalently, this occurs if f , g , and h all have continuous first derivatives that do not all vanish simultaneously. Since $\mathbf{r}'(t) \neq \vec{0}$, the particle never stops or reverses direction.

Now for some physics applications:

Definitions. If $\mathbf{r}(t)$ is the position vector of a particle moving along a smooth curve in space, then

$$\mathbf{v}(t) = \mathbf{r}'(t)$$

is the particle's **velocity vector**, tangent to the curve. It always points in the direction of motion. Additionally,

- The speed of the particle at point t_0 is given by $\|\mathbf{v}(t_0)\| = \|\mathbf{r}'(t_0)\|$
- The particles acceleration is given by $\mathbf{a}(t) = \mathbf{v}'(t) = \mathbf{r}''(t)$.
- The unit vector $\frac{\mathbf{v}}{\|\mathbf{v}\|}$ is the direction of movement at time t .

Example. Find the velocity, speed, and acceleration of a particle whose motion in space is given by the position vector

$$\mathbf{r}(t) = \langle 2 \cos(t), 2 \sin(t), 5 \cos^2(t) \rangle.$$

Since the derivative is defined to be a limit component-wise, the derivative works the same. There are only a few rules that are worth mentioning explicitly:

$$(i) \quad \frac{d}{dt}[\mathbf{u}(t) \cdot \mathbf{v}(t)] = \mathbf{u}'(t) \cdot \mathbf{v}(t) + \mathbf{u}(t) \cdot \mathbf{v}'(t)$$

$$(ii) \quad \frac{d}{dt}[\mathbf{u}(t) \times \mathbf{v}(t)] = \mathbf{u}'(t) \times \mathbf{v}(t) + \mathbf{u}(t) \times \mathbf{v}'(t)$$

$$(iii) \quad \frac{d}{dt}[\mathbf{u}(f(t))] = f'(t)\mathbf{u}'(f(t))$$

The tangent line to a smooth curve $\mathbf{r}(t) = \langle f(t), g(t), h(t) \rangle$ at $t = t_0$ is the line that passes through the point $(f(t_0), g(t_0), h(t_0))$ parallel to $\mathbf{v}(t_0)$. Thus, we can find parametric equations for this line:

Example. Find parametric equations for the line tangent to $\mathbf{r}(t) = \langle 2 \sin(t), t^4 - 6 \cos(t), 3e^{4t} \rangle$ at $t_0 = 0$.

Lastly, we make the following observation: If your particle's length-vector is always the same (this occurs, for example, when we consider a particle that strictly moves around the unit-sphere),

then we have

$$\mathbf{r}(t) \cdot \mathbf{r}(t) = \|\mathbf{r}(t)\|^2 = c^2.$$

But if we differentiate both sides and apply the product rule, this becomes

$$\mathbf{r}'(t) \cdot \mathbf{r}(t) + \mathbf{r}(t) \cdot \mathbf{r}'(t) = 2\mathbf{r}'(t) \cdot \mathbf{r}(t) = 0$$

and therefore we reach the following conclusion:

Theorem. If $\mathbf{r}(t)$ is a differentiable vector-function of constant length, then

$$\mathbf{r}(t) \cdot \mathbf{r}'(t) = 0$$

and therefore the position of the particle and the velocity of the particle are orthogonal.